2110203: COMP ENG MATH II (2026/1) Homework 2

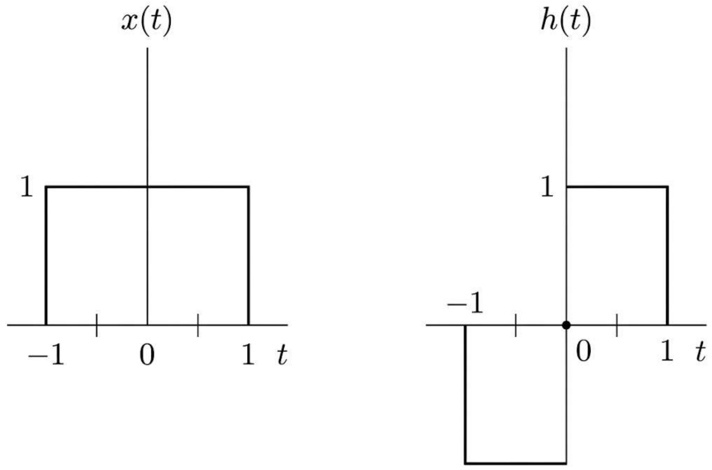
Name: Student ID:

Write solutions and give explanations (1 point)

SEND ONLY PROBLEM 1. and 2.

1. (0.6 points, 0.15 points each) Determine the convolution $y(t) = h(t) * x(t)$ using <u>Graphical Interpretation</u> of the pairs of the signals shown

   1.1.



Final answer: $y(t) =$

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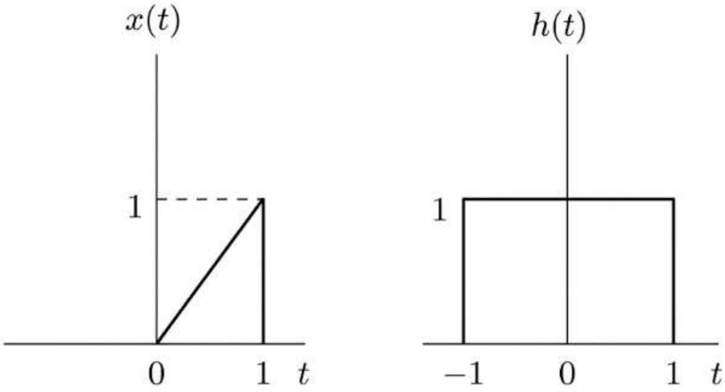
1.2.



Final answer: $y(t) =$

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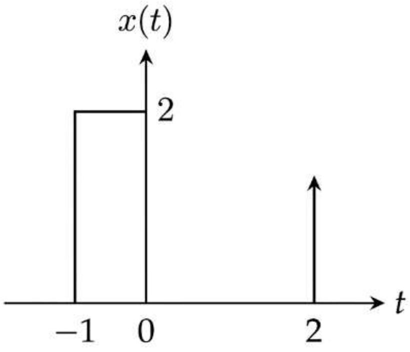
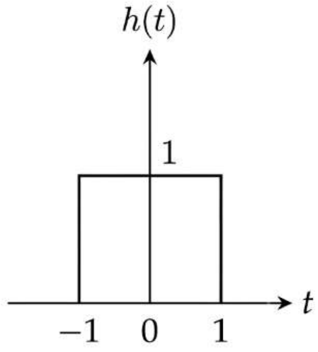
1.3.





Final answer: $y(t) =$

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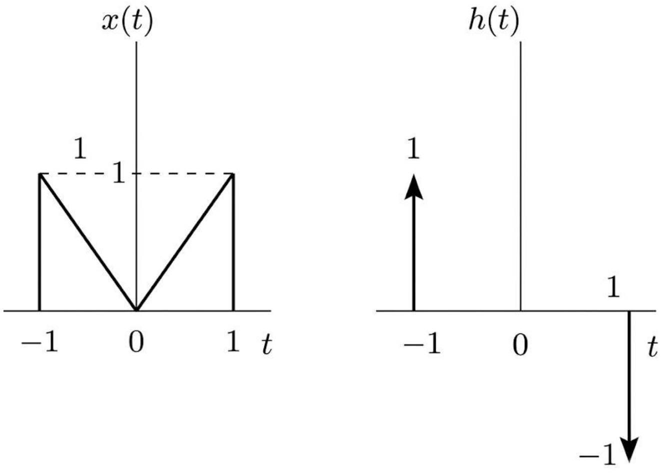
1.4.



Final answer: $y(t) =$

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2. (0.4 points, 0.1 points each) Find the convolution $y[n] = h[n] * x[n]$ of the following signals

2.1.

$$x[n] = \begin{cases} -1 \,; & -5 \le n \le -1 \\ 1 \,; & 0 \le n \le 4 \end{cases}$$
$$h[n] = 2u[n]$$

2.2.

$$x[n] = \left(\frac{1}{2}\right)^n u[n]$$
$$h[n] = \delta[n] + \delta[n-1] + \left(\frac{1}{3}\right)^n u[n]$$

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2.3.

$$x[n] = u[n]$$
$$h[n] = 1 \ ; 0 \leq n \leq 9$$

2.4.

$$x[n] = \left(\frac{1}{3}\right)^n u[n]$$
$$h[n] = \delta[n] + \left(\frac{1}{2}\right)^n u[n]$$

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THE QUESTIONS BEYOND THIS ARE OPTIONAL, PRACTICE ONLY.

3.

$$\text{Your Student ID: } \_\ \_\ \_\ \_\ \underset{A}{\_}\ \underset{B}{\_}\ \underset{C}{\_}\ \underset{D}{\_}\ \underset{E}{\_}\ \_$$

(Substitute the digits A, B, C, D, E from your ID into the parameters below)

Consider two continuous-time signals, $x(t)$ and $v(t)$, defined as follows:

$$x(t) = \frac{10}{A+B+2}[u(t-C) - u(t-(C+A+B+2)]$$
$$v(t) = 2[u(t-D) - u(t-(D+A+B+2)]$$

A system processes and input signal $z(t)$ to produce an output $y(t)$ according to the equation:

$$y(t) = (t-E)z(t)$$

3.1. Find the continuous-time convolution $z(t) = x(t) * v(t)$ using the graphical interpretation method. Express your final answer for $z(t)$ as a piecewise function.

3.2. Determine whether the system described by $y(t) = (t-E)z(t)$ is a Time-varying or Time-invariant system.

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4. Evaluate the convolution of the following signals

   4.1. $\mathrm{rect}(\frac{t-a}{a}) * \delta(t-b)$

   4.2. $\mathrm{rect}(\frac{t}{a}) * \mathrm{rect}(\frac{t}{a})$

   4.3. $t[u(t)-u(t-1)] * u(t)$

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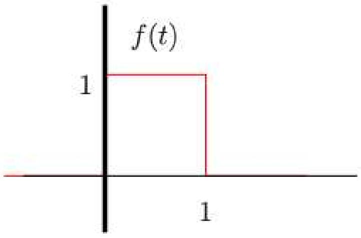
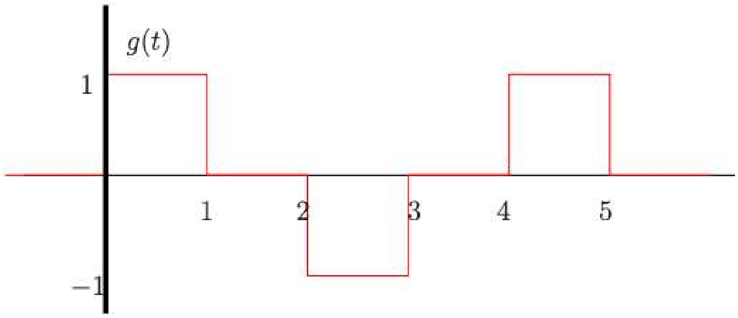
5. Let $f(t)$ and $g(t)$ be given as follows:





5.1. Sketch the function $x(t) = f(t) * g(t)$

5.2. Show that if $a(t) = b(t) * c(t)$, then $\big(Mb(t)\big) * c(t) = Ma(t)$, for any real number M (Hint: use the convolution integral formula)

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6. Find the convolution $y[n] = h[n] * x[n]$ of the following signals

6.1. $x[n] = \{1, -\frac{1}{2}, \frac{1}{4}, -\frac{1}{8}, \frac{1}{16}\}, h[n] = \{1, -1, 1, -1\}$

6.2. $x[n] = \{1, 2, 3, 0, -1\}, h[n] = \{2, -1, 3, 1, -2\}$

6.3. $x[n] = \{3, \frac{1}{2}, -\frac{1}{4}, 1, 4\}, h[n] = \{2, -1, \frac{1}{2}, -\frac{1}{2}\}$

6.4. $x[n] = \{-1, \frac{1}{2}, \frac{3}{4}, -\frac{1}{5}, 1\}, h[n] = \{1, 1, 1, 1, 1\}$

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7. Are the following systems linear or time invariant?
   7.1. $x(t) \rightarrow System(a) \rightarrow 7x(t-1)$
   7.2. $x(t) \rightarrow System(b) \rightarrow \cos\big(2x(t)\big)$
   7.3. $x(t) \rightarrow System(c) \rightarrow t$
   7.4. $x(t) \rightarrow System(d) \rightarrow x(t) + t$

If you want to try the coding problem, use this link.
https://colab.research.google.com/drive/1igo0KyyNxyw7ZnCvEwt92wiRUWwJPYMD?usp=sharing